

The missing move changes everything

Name: _____

Date: _____

Two players alternate. Take **1, 3, or 4** counters each turn. Taking 2 is illegal. Never take more than remain. Taking the last counter wins.

1. Play and suspect.

Try piles of 5, 6, and 7. Trade who goes first. A win gives you a clue; it does not yet prove a strategy.

2. Solve small positions.

L means the player about to move loses against perfect replies. W means that player has a winning strategy. Zero is L: the previous player took the last counter.

0	1	2	3	4	5	6
7	8	9	10	11	12	13

3. Justify each mark.

To label W, draw one legal move to an L. To label L, check **every** legal move and show it leads to a W. Start with 0 and work upward.

One W with its winning move; one L with all its replies:

Build first. Explain by pointing, talking, or drawing.

One good move, or every bad move?

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4. Two claims to test.

Alex says: “From 8, take 3 because taking more is better.” Bea says: “From 10, there is only one winning move.” Use your chart to check both.

Alex’s move and a better move, with reasons:

Every winning move from 10:

5. Make a strategy card.

Choose a pile between 8 and 14. If it is W, show a winning first move and how you will use your chart after that. If it is L, explain why every first move gives away the advantage.

My pile and my strategy certificate:

Go further: Mark 14 through 20. Do you see a repeating pattern? What would you need to prove before trusting it for a pile of 100?

Build first. Explain by pointing, talking, or drawing.